

Project Goal

Automatic detection of anomalous login events from logs in order to identify early signs of Brute Force and Credential Stuffing attacks.

Methodology

Generate a synthetic dataset, extract features, train an Isolation Forest model, evaluate performance, and rank suspicious events.

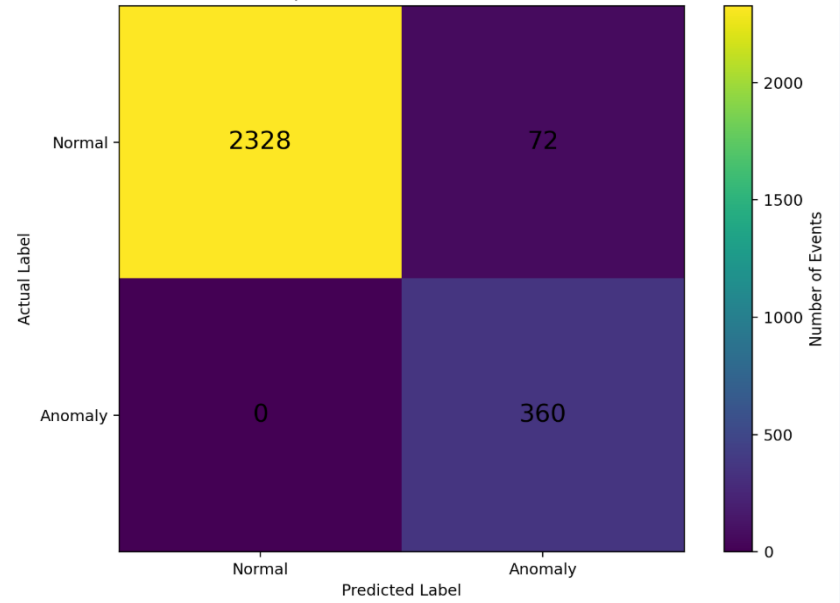
Cybersecurity Value

Helps security teams prioritize suspicious events and investigate them before they develop into a significant security incident.

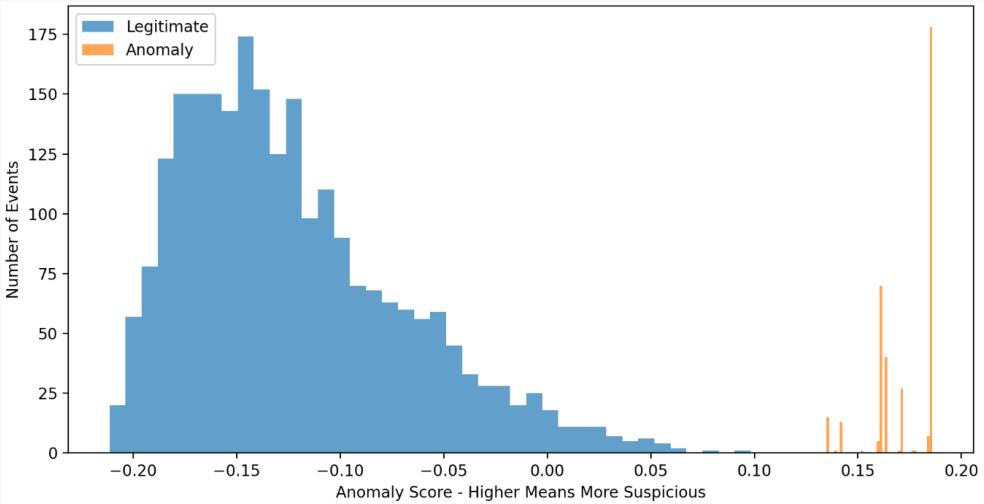
Key Features

Failed login attempts, number of different usernames from the same IP, login hour, geographic risk, IP reputation, attempt rate, success ratio, and new device detection.

Confusion matrix



Anomaly Score Distribution



Results

Accuracy=0.974 | Precision=0.833
Recall=1.000 | F1=0.909

System Architecture

Login Logs → Feature Extraction → Isolation Forest → Suspicious Event Ranking

Future Work

Connect to real logs, build a real-time dashboard, integrate with a SIEM, and add an automatic explanation for each alert.